

Report 6 - Shaft Model Family for OpenHPL/OpenHPLjl

Madhusudhan Pandey

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Abstract

This report introduces the shaft model family as the mechanical bridge between turbine torque and generator torque. The first implementation is a lumped single-inertia shaft with viscous damping and an acausal rotational connector.

1 Updated Context

The shaft transfers turbine mechanical power to the generator:

$$\boxed{\text{Turbine} \rightarrow \text{Shaft} \rightarrow \text{Generator}}.$$

Its first role in OpenHPLjl is to represent rotational inertia and speed dynamics.

2 Generalized Concept Model

The shaft receives drive and load torques and exposes one common angular speed:

$$\tau_{drive}, \tau_{load} \rightarrow J \rightarrow \omega.$$

3 Other Models Available in the Literature

Useful model levels are

$$\boxed{\text{rigid massless shaft} \rightarrow \text{single inertia} \rightarrow \text{two-mass torsional shaft} \rightarrow \text{distributed flexible shaft}}.$$

4 Mathematics Involved

For the lumped shaft,

$$\boxed{J\dot{\omega} = \tau_{drive} + \tau_{load} - D(\omega - \omega_0)}.$$

Mechanical power is

$$P_m = \tau\omega.$$

5 OpenHPL-Specific Modeling Interpretation

OpenHPL includes rotational mechanical connections between turbine and generator. OpenHPLjl mirrors that physical interface with an acausal rotational port containing angular speed and flow-conserved torque.

6 Implementation in Julia ModelingToolkit / SciML

```
@connector RotationalPort begin
    omega(t)
    tau(t), [connect = Flow]
end

@component function LumpedShaft(; name,
    J=1e5, damping=0.0, omega0=2*pi*50)
    ...
    J * D(omega) ~ drive.tau + load.tau -
                    damping*(omega-omega0)
end
```

7 Model Assembly and Connections

The shaft is intended to connect the turbine mechanical port to the generator mechanical port. The next package step will give the turbine an explicit rotational connector.

8 Numerical Experiment / Validation

For

$$J = 100, \quad \tau_{drive} = 20, \quad \tau_{load} = -10,$$

the acceleration is

$$\dot{\omega} = \frac{10}{100} = 0.1 \text{ rad s}^{-2}.$$

Starting at 100 rad s^{-1} ,

$$\boxed{\omega(2) = 100.2 \text{ rad s}^{-1}}.$$

9 Expected Outputs

Key outputs are

$$\omega(t), \quad \dot{\omega}(t), \quad \tau_{drive}(t), \quad \tau_{load}(t).$$

10 Engineering Interpretation

The shaft introduces mechanical inertia, the central storage mechanism governing electromechanical frequency dynamics.

Conclusion

The first shaft model is intentionally minimal. Next steps are a two-mass torsional model and explicit coupling to turbine torque and generator electromagnetic torque.